

Transport in animals

**Circulatory system:** A system of blood vessels with a pump and valves to ensure one way flow of blood

**Single circulation:** When the blood goes through the heart once for each complete circulation

**Double circulation:** When the blood passes through the heart twice for each complete circulation.

**Advantages of double circulation:**

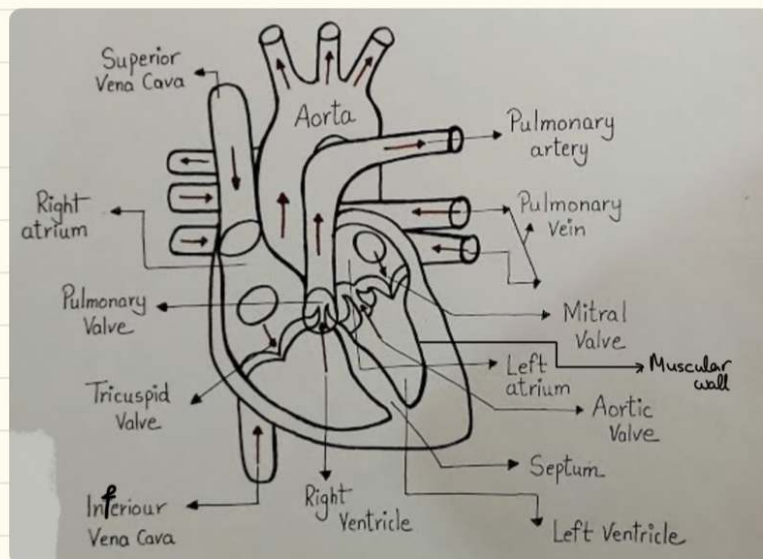
- 1- Maintaining a high blood pressure to all the major organs of the body
- 2- Separates the oxygenated blood from the deoxygenated blood to allow the body cells to receive sufficient oxygen
- 3- Allows the systemic circulation to have a high blood pressure since blood will be pumped to the rest of the body and will need to travel over long distances.

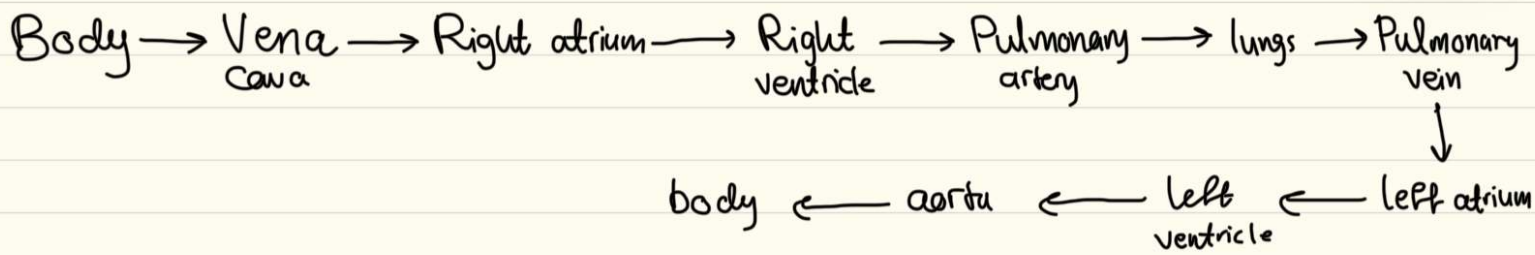
Fish is the only living organism that has single circulation

Mammals, birds & reptiles have double circulation

**Pulmonary circulation** carries deoxygenated blood from the heart to the lungs through the pulmonary artery. Then returns oxygenated blood from the lungs to the heart through the pulmonary vein.

**Systemic circulation** involves pumping oxygenated blood from the heart to the rest of the body through the aorta, and returning deoxygenated blood from the body to the heart through the venacava





Values: Prevent the backflow of blood

1-Atrioventricular valves: between the atria and ventricles

\* Right side: Tricuspid valve

\* Left side: bicuspid/mitral valve

2-Semi-Lunar valves: between ventricles and vessels

\* Pulmonary valve at the pulmonary artery

\* Aortic valve at the aorta

\* The heart is made of a special type of muscle called cardiac muscle.

1-The right atrium collects deoxygenated blood and pumps it to the right ventricle.

2- Right ventricle pumps the deoxygenated blood to the lungs through pulmonary artery

3- Pulmonary artery carries deoxygenated blood from the right ventricle to the lungs

4- Pulmonary vein carries oxygenated blood from the lungs to the left atrium.

5- Left atrium collects oxygenated blood and pumps it to the left ventricle

6- Left ventricle pumps oxygenated blood from the left ventricle to the rest of the body through the aorta.

7- Aorta carries oxygenated blood from the left ventricle to the body

\* Atrioventricular valves: Prevent the backflow of blood from ventricles to the atria when ventricles contract, as the pressure will be lower in the atria.

\* Semi-lunar valves: Prevents the backflow of blood from the arteries into ventricles.

**Septum:** A muscular wall that separates the right side of the heart from the left side. It is important because it separates the oxygenated blood from the deoxygenated blood.

\* The atria have much thinner walls than the ventricles because they receive blood rather than pumping it.

\* The left ventricle has a wall three times thicker than the right ventricle. This is because the right ventricle only needs enough pressure to pump the blood to the lungs, which are next to the heart. However, the left ventricle pumps the blood to all the body organs.

## Cardiac Cycle

\* **Cardiac diastole:** Atria and ventricles are relaxed, volume of the atria and ventricles increases and blood pressure decreases. Blood flows into the heart. Semilunar valves are closed to prevent the flow of blood into arteries. Atrioventricular valves are opened.

\* **Atrial systole:** Muscles in the walls of the atria contract, decreasing the volume and increasing the blood pressure; pushing the blood through the opened atrioventricular valves, which are forced open due to high pressure, and into the ventricles. Semilunar valves are closed.

\* **Ventricular systole:** Atria relax, muscles in the walls of ventricles contract, decreasing the volume and increasing the blood pressure. Blood is forced into the arteries. Semi-lunar valves are open due to high pressure. Atrioventricular valves are closed to prevent the backflow of blood to atria. Semi-lunar valves will shut once the blood gets into the arteries to prevent the backflow of blood into ventricles.

The activity of the heart can be monitored by: 1- ECG

2- Measuring pulse rate

3- Listening to the sounds of the valves closing.

## Heart beat

- Normally, a heart beats 60 to 75 times per minute

\* Pulse represents the expansion and relaxation of the arteries as a result of the blood being pumped through them.

\* During exercise, heart rate increases. This is because muscles need more energy so the heart has to deliver more oxygen and glucose for respiration

\* Heart beat is controlled by the pacemaker in the right atrium

## Coronary heart disease

CHD is the blockage of coronary arteries

Coronary arteries supply the heart with nutrients. When it gets blocked, the heart muscle becomes deprived of oxygen and glucose, and toxic wastes like lactic acid build up in the heart. Cells become damaged and part of the heart muscle stops contracting → causing heart attack.

1- Cholesterol deposits in the blood vessels forming a plaque

2- Making the lumen of the artery narrower and the artery becomes stiffer, which increases blood pressure

3- Endothelial lining of the artery becomes damaged and blood clots form.

4- The blood clots stop the blood from getting through, forming a thrombosis.

\* The person suffers from heart attacks or might suffer a stroke if the vessel is supplying the brain. This is because cells will not receive any blood containing oxygen and glucose, so they will not be able to carry out aerobic respiration to release the energy needed for contraction, so the heart cells or brain cells die

## Possible risk factors of coronary heart disease:

- 1-Diet: an unbalanced diet with too much animal fat and too many calories lead to obesity. Being overweight puts extra strain on the heart, which makes it more difficult for the person to exercise. The fat deposit in arteries, restricting the blood flow, and cause blood clots → leading to a heart attack.
- 2-Stress: Emotional stress often leads to raised blood pressure. High blood pressure may increase the rate at which atheroma (fat's deposit) are formed in arteries.
- 3-Smoking: Smokers are two to three times more likely to die from heart attacks than non-smokers at similar age. The carbon monoxide and other chemicals in cigarette smoke damages the lining of the arteries, allowing atheroma to form.
- 4-Genetics: Coronary heart disease appears to be passed from generation to the next generation.
- 5-Age & sex: When a person gets older, the risk from suffering from coronary heart disease increases. Males are more at risk of a heart attack than females. The hormone oestrogen offers females some protection from heart disease until later in life, when oestrogen levels drop. Males also tend to have less healthy lifestyles than females.
- 6-Lack of exercise: Heart muscle loses its tone and becomes less efficient at pumping blood when we do not exercise. Lack of exercise results in slower blood flow, which may allow atheroma to form.

## Treating CHD

- 1-Drugs that lower cholesterol levels in blood like statins reduce the risk of blood clots like aspirin or beta blockers which decrease blood pressure.
- 2-Angioplasty: Inflating the collapsed artery by a balloon which gets filled with water
- 3-Inserting a stent in the collapsed artery to keep it open
- 4-Replace damaged artery in coronary bypass surgery
- 5-Heart transplant

# Blood vessels

## Arteries:

Function: Transport oxygenated blood at high pressure from the heart to the rest of the body. Except for the pulmonary artery: carries deoxygenated blood from the heart to the lungs.

## Adaptations:

- Thick muscular walls to withstand high blood pressure.
- Elastic walls that stretch and recoil when blood is pushed in or out through it.
- Narrow lumen to maintain high blood pressure.
- Valves are absent because high pressure prevents the backflow of blood.

## Veins:

Function: Transport deoxygenated blood from the rest of the body towards the heart. Except for the pulmonary vein; carries oxygenated blood from the lungs to the heart.

## Adaptations:

- Thin walls because they carry blood at low pressure.
- Wide lumen to reduce resistance to blood flow.
- Have valves to prevent the backflow of blood because it is pumped at low pressure.

\* Veins are present in between muscles to help the blood move by contraction.

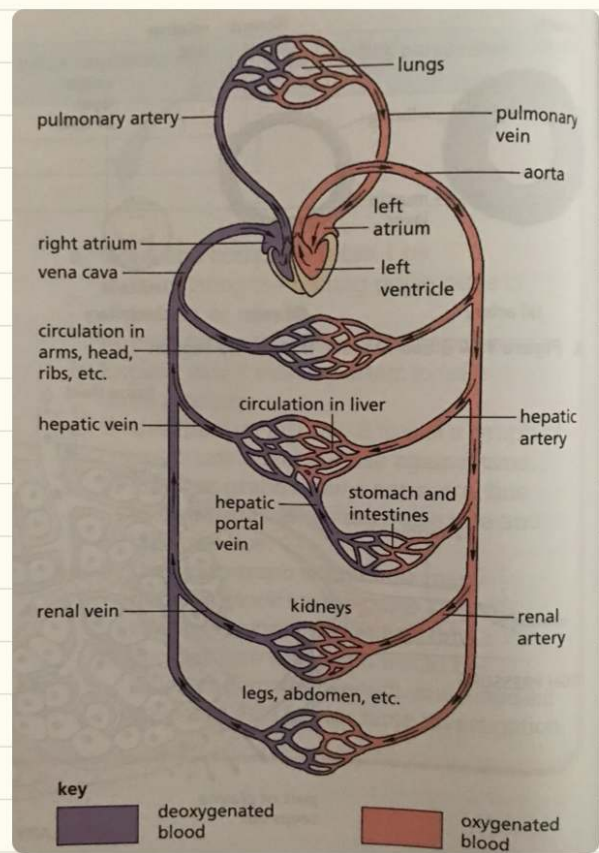
## Capillaries:

Functions: Allow the exchange of substances between blood and tissues such that useful substance, glucose and oxygen, diffuse into cells. And waste/toxic substances, urea and  $\text{CO}_2$ , diffuse out of cells.

## Adaptations:

- 1- Walls are one cell thick and permeable to allow the diffusion of materials between the capillaries and surrounding tissues.
- 2- Lumen is one red blood cell thick as white blood cells can squeeze through.
- 3- Highly branched to increase surface area for diffusion.
- 4- Present in very large numbers.
- 5- Valves are absent because blood is still under pressure.

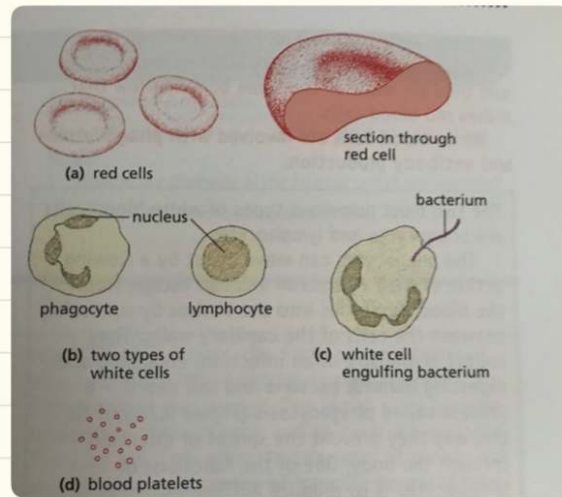
Pulmonary: Lungs  
Renal: Kidney  
Hepatic: Liver



## Liver

Hepatic vein carry deoxygenated blood from the liver to the heart  
Hepatic artery carry oxygenated blood from the heart to the liver  
Hepatic portal vein transports nutrients such as amino acids, glucose, amino acids and vitamins to the liver from the small intestine.

# Contents of the blood



**Red blood cells:** Transport oxygen around the body

**Adaptations:**

- 1- Have a biconcave shape to squeeze into blood capillaries and to provide a large surface area for carrying more oxygen
- 2- Have no nucleus to have more space to fit more hemoglobin and carry more oxygen
- 3- Have red protein hemoglobin which binds and carries oxygen

**White blood cells:** Phagocytosis and antibody production.

- Phagocytes encounter a bacteria and recognize the antigens on the bacteria as foreign antigens. They engulf the bacteria into a vacuole, and secrete enzymes that destroy the bacteria. The destroyed bacteria is then released.
- Lymphocytes release antibodies. / antibody production

**Plasma:** The transport of blood cells, ions, nutrients, urea, hormones & carbon dioxide.

**Platelets:** Blood clotting

\* Blood vessels are damaged → which activates the platelets. A chain reaction occurs which activates thrombin, a protease enzyme. Thrombin converts the soluble protein fibrinogen into the insoluble protein fibrin. Fibrin forms a mesh that traps red blood cells and platelets which stick together forming a plug that seals the wound.

**Advantages of blood clotting:**

- Prevents blood loss
- Prevents pathogens from entering the body